A Thermodynamic Extension of the Interest Rate Functional Representation (IRFR) Phase 2–4 Framework

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Abstract

This document develops the thermodynamic structure implied by the Instantaneous Rolling Forward Rate (IRFR), centred on two key invariants: the Work–to–Mean (WTM) and Entropy–to–Mean (ETM) functionals. Phase 2 introduces these invariants and shows that they fully characterise non-equilibrium IRFR dynamics. Phase 3 incorporates persistent central-bank intervention, showing how policy forcing distorts drift while preserving the thermodynamic invariants. Phase 4 introduces metastability, demonstrating how suppressed drift leads to exponential accumulation of hidden macroeconomic energy and eventual catastrophic failure when a critical threshold is breached.

1 Thermodynamic Foundations of IRFR

1.1 The Work-to-Mean Functional (WTM)

Let $x_{\text{obs}}(t,T)$ denote the observed yield curve and $x_e(T-t)$ the equilibrium IRFR attractor.

Define the maturity-integrated deviation:

$$A_{\text{WTM}}(t) = \int_0^\infty \left[x_{\text{obs}}(t,T) - x_e(T-t)\right] dT. \tag{1}$$

Interpretation. $A_{\text{WTM}}(t)$ measures the aggregate "macroeconomic energy" stored in the deviation of the yield curve from equilibrium. It is the maturity-integrated mean-reversion work the system must exert to return to the IRFR attractor.

1.2 WTM Identification and Amplitude Reconstruction

The latent non-equilibrium IRFR curve has the constrained form

$$x_{\text{neq}}(t,T) = x_\infty + (x_{t,t} - x_\infty)\, e^{-a^2(T-t)}. \tag{2}$$

The equilibrium IRFR curve is

$$x_e(T-t) = x_\infty \left(1 - \tfrac{1}{2} e^{-a^2(T-t)}\right). \tag{3}$$

The Work–to–Mean functional is therefore

$$A_{\mathrm{WTM}}(t) = \int_0^\infty \big(x_{\mathrm{neq}}(t,T) - x_e(T-t)\big)\, dT. \tag{4}$$

Step 1: Compute the deviation. Substituting (2)–(3),

$$x_{\mathrm{neq}}(t,T) - x_e(T-t) = \left(x_{t,t} - x_\infty + \tfrac{1}{2}x_\infty\right) e^{-a^2(T-t)} = \left(x_{t,t} - \tfrac{1}{2}x_\infty\right) e^{-a^2(T-t)}.$$

Step 2: Integrate over maturity. Let $\tau = T - t$. Then

$$A_{\mathrm{WTM}}(t) = \left(x_{t,t} - \tfrac{1}{2}x_\infty\right) \int_0^\infty e^{-a^2\tau}\, d\tau = \left(x_{t,t} - \tfrac{1}{2}x_\infty\right) \frac{1}{a^2}.$$

Step 3: Solve for the IRFR amplitude. Rearranging,

$$x_{t,t} - \tfrac{1}{2}x_\infty = a^2 A_{\mathrm{WTM}}(t),$$

so the short-maturity IRFR is

$$\boxed{x_{t,t} = \tfrac{1}{2}x_\infty + a^2 A_{\mathrm{WTM}}(t).} \tag{5}$$

Step 4: Reconstruct the full IRFR curve. Substituting (5) into (2) gives

$$\boxed{x_{\mathrm{neq}}(t,T) = x_\infty + \left(a^2 A_{\mathrm{WTM}}(t) - \tfrac{1}{2}x_\infty\right) e^{-a^2(T-t)}.} \tag{6}$$

This is the unique non-equilibrium IRFR curve consistent with the observed WTM functional.

1.3 Entropy Matching Hypothesis

In Section 1.2 we showed that the Work–to–Mean functional (WTM) determines the unique non-equilibrium IRFR curve compatible with the observed maturity-integrated deviation from the equilibrium attractor. We now introduce a second thermodynamic quantity: the Entropy-to-Mean functional (ETM), which encapsulates the total fluctuation intensity embedded in the observed yield curve.

Definition (Observed ETM). Let $\sigma_{\mathrm{obs}}(t,T)$ denote the instantaneous volatility associated with the forward rate of maturity T. Define the observed entropy functional:

$$E^{\mathrm{obs}}_{\mathrm{ETM}}(t) = \int_0^\infty \sigma_{\mathrm{obs}}(t,T)\, dT. \tag{7}$$

The ETM functional measures the maturity-integrated fluctuation activity of the system and plays the role of a thermodynamic entropy measure.

The Entropy-Matching Hypothesis. We now state the central thermodynamic requirement of the IRFR framework:

$$\boxed{E^{\text{obs}}_{\text{ETM}}(t) = E^{\text{neq}}_{\text{ETM}}(t)} \tag{8}$$

That is, *the observed entropy of the yield curve must equal the entropy implied by the latent non-equilibrium IRFR curve reconstructed from the WTM functional.*

Interpretation.

- WTM determines the total "energy" stored in the deviation of the yield curve from the equilibrium IRFR attractor.
- ETM measures the aggregate volatility (fluctuation intensity) present across maturities.
- The entropy-matching hypothesis asserts that the observed volatility structure reflects the same underlying thermodynamic displacement encoded by the WTM functional.

In other words, the "energy" (WTM) of the deviation and the "entropy" (ETM) of the fluctuations must correspond to a single underlying non-equilibrium IRFR state. This ensures that the non-equilibrium IRFR curve recovered from WTM is not merely geometrically admissible but thermodynamically consistent.

Consequences. Equation (8) implies that any persistent discrepancy between the observed volatility and the volatility implied by the reconstructed IRFR curve must arise from an additional forcing mechanism—most naturally interpreted as policy intervention. This observation forms the conceptual bridge to Phase 3.

1.4 Evolution of the WTM Functional and the Thermodynamic Decay Law

We begin from the IRFR manifold identity, which holds at every time:

$$x(t,T) - x_e(T-t) = \big(x(t,t) - \tfrac{1}{2}x_\infty\big)\, e^{-a^2(T-t)}. \tag{9}$$

Our objective is to compute the deviation at the next time step $t + \Delta t$, integrate it over maturity, and thereby obtain the updated Work–to–Mean functional $A_{\text{WTM}}(t + \Delta t)$.

Step 1: Forward evolution of the short rate. At zero maturity, the IRFR drift satisfies

$$\frac{d}{dt}x(t,t) = -a^2\big(x(t,t) - \tfrac{1}{2}x_\infty\big).$$

A first-order (forward Euler) step gives:

$$x(t+\Delta t, t+\Delta t) = x(t,t) - 2a^2 x(t,t)\,\Delta t + a^2 x_\infty\,\Delta t. \tag{10}$$

Step 2: Deviation at time $t + \Delta t$. Applying (9) with $t \mapsto t + \Delta t$ and defining $\theta := T - t - \Delta t$, we obtain

$$x(t + \Delta t, T) - x_e(\theta) = \Big(x(t,t) - 2a^2 x(t,t)\,\Delta t + a^2 x_\infty\,\Delta t - \tfrac{1}{2}x_\infty\Big)e^{-a^2\theta}. \tag{11}$$

Step 3: Integration over maturity. The WTM functional at the next step satisfies

$$A_{\mathrm{WTM}}(t + \Delta t) = \int_0^\infty \big[x(t + \Delta t, T) - x_e(\theta)\big]\,d\theta.$$

Substituting (11) and using $\int_0^\infty e^{-a^2\theta}d\theta = 1/a^2$ yields:

$$A_{\mathrm{WTM}}(t + \Delta t) = \frac{x(t,t) - 2a^2 x(t,t)\,\Delta t + a^2 x_\infty\,\Delta t - \frac{1}{2}x_\infty}{a^2}. \tag{12}$$

Step 4: Expressing the result in terms of $A_{\mathrm{WTM}}(t)$. By definition,

$$A_{\mathrm{WTM}}(t) = \frac{x(t,t) - \frac{1}{2}x_\infty}{a^2}, \qquad \Rightarrow \qquad x(t,t) - \tfrac{1}{2}x_\infty = a^2 A_{\mathrm{WTM}}(t). \tag{13}$$

Substituting (13) into (**??**) and simplifying gives:

$$A_{\mathrm{WTM}}(t + \Delta t) = A_{\mathrm{WTM}}(t)\big(1 - a^2\Delta t\big) + O(\Delta t^2). \tag{14}$$

Step 5: The thermodynamic decay law. Taking $\Delta t \to 0$ in (14) yields the continuous-time evolution equation:

$$\frac{d}{dt}A_{\mathrm{WTM}}(t) = -\,a^2 A_{\mathrm{WTM}}(t),$$

whose solution is the exponential relaxation law:

$$\boxed{A_{\mathrm{WTM}}(t + \Delta t) = A_{\mathrm{WTM}}(t)\,e^{-a^2\Delta t}.} \tag{15}$$

Equation (15) is the *thermodynamic decay law* of the IRFR system: the Work–to–Mean functional dissipates exponentially at the intrinsic relaxation rate a^2 dictated by the IRFR drift. This identifies $A_{\mathrm{WTM}}(t)$ as the natural thermodynamic energy variable of the model.

1.5 Volatility Preservation at Time $t + \Delta t$

In this section we show that both the IRFR volatility and the observed volatility are preserved across a time step $t \mapsto t + \Delta t$. Although the deviation from equilibrium shrinks under IRFR drift, the fluctuation intensity (volatility) remains unchanged. This is the thermodynamic entropy-preservation property of the IRFR system.

IRFR volatility. Under the identified equilibrium structure, the diffusion coefficient of the IRFR satisfies

$$g(t,T) = a\big(x(t,T) - x_\infty\big) - a x_\infty e^{-a^2(T-t)}, \tag{16}$$

so the instantaneous volatility is

$$\sigma_{\mathrm{IRFR}}(t,T) = \sqrt{2}\, g(t,T).$$

Using the IRFR manifold representation,

$$x(t,T) = x_\infty + \big(x(t,t) - \tfrac{1}{2}x_\infty\big)e^{-a^2(T-t)},$$

we obtain

$$g_{\mathrm{IRFR}}(t,T) = a\big(x(t,t) - \tfrac{3}{2}x_\infty\big)\, e^{-a^2(T-t)}. \tag{17}$$

At the next time step $t + \Delta t$, the short-maturity forward rate evolves as

$$x(t+\Delta t, t+\Delta t) = x(t,t) - 2a^2 x(t,t)\Delta t + a^2 x_\infty \Delta t. \tag{18}$$

Substituting (18) into (17), and noting that

$$e^{-a^2(T-t-\Delta t)} = e^{a^2\Delta t}\, e^{-a^2(T-t)},$$

one finds that the $O(\Delta t)$ terms cancel, giving

$$g_{\mathrm{IRFR}}(t+\Delta t, T) = g_{\mathrm{IRFR}}(t,T), \tag{19}$$

and hence

$$\boxed{\sigma_{\mathrm{IRFR}}(t+\Delta t, T) = \sigma_{\mathrm{IRFR}}(t,T).} \tag{20}$$

Observed volatility. Let the observed curve be decomposed into the IRFR component and a distortion layer:

$$x_{\mathrm{obs}}(t,T) = x_{\mathrm{IRFR}}(t,T) + \Pi(t,T).$$

In the absence of new policy forcing over the interval $[t, t+\Delta t]$, the distortion satisfies the thermodynamic decay law

$$\Pi(t+\Delta t, T) = \Pi(t,T)e^{-a^2\Delta t}. \tag{21}$$

Since $\sigma_\Pi(t,T) \propto \Pi(t,T)$, combining (21) with the identity

$$e^{-a^2(T-t-\Delta t)} = e^{a^2\Delta t}\, e^{-a^2(T-t)},$$

shows that the exponential factors cancel, yielding

$$\sigma_\Pi(t+\Delta t, T) = \sigma_\Pi(t,T).$$

Because IRFR volatility is preserved by (20), the total observed volatility satisfies

$$\boxed{\sigma_{\mathrm{obs}}(t+\Delta t, T) = \sigma_{\mathrm{obs}}(t,T).} \tag{22}$$

Interpretation. Equation (20) shows that IRFR volatility is time-invariant under natural relaxation. Equation (22) shows that observed volatility is also invariant in the absence of new forcing. Thus, although the yield curve deviation from equilibrium decays at rate a^2, the associated fluctuation intensity remains unchanged. This is the entropy-preservation law of the IRFR thermodynamic system.

1.6 Policy Forcing at the Short End and Propagation Along the IRFR Manifold

In the absence of intervention, the IRFR system evolves according to the thermodynamic drift law derived in Sections 1.2–1.4, which determines the expected IRFR curve at time $t + \Delta t$:

$$\mathbb{E}\big[x_{\mathrm{IRFR}}(t+\Delta t, T)\big] = x_\infty + \big(a^2 A_{\mathrm{WTM}}(t) - \tfrac{1}{2}x_\infty\big)\, e^{-a^2(T-t-\Delta t)}. \tag{23}$$

We now introduce a short-rate intervention. Suppose the central bank enforces a short rate at time $t + \Delta t$ equal to

$$x_{\mathrm{obs}}(t+\Delta t, t+\Delta t),$$

which differs from the natural IRFR expectation $\mathbb{E}[x_{\mathrm{IRFR}}(t+\Delta t, t+\Delta t)]$. Define the short-rate policy shock:

$$\Delta x_{t+\Delta t} = x_{\mathrm{obs}}(t+\Delta t, t+\Delta t) - \mathbb{E}[x_{\mathrm{IRFR}}(t+\Delta t, t+\Delta t)]\,. \tag{24}$$

The WTM interpretation of the short-rate shock. Since the Work–to–Mean functional measures maturity-integrated deviation from equilibrium, a short-rate shock over an interval Δt injects the WTM area

$$\Delta A_{\mathrm{WTM}}(t+\Delta t) = \Delta x_{t+\Delta t}\,\Delta t. \tag{25}$$

This represents the macroeconomic "energy" added to the system by the intervention.

Thermodynamic propagation of the injected area. Because the IRFR system lies on a one-dimensional exponential manifold, any deviation at the short end must propagate along the same kernel $e^{-a^2(T-t-\Delta t)}$. Thus the deformation to the forward curve at $t+\Delta t$ has the form

$$\Delta x_{t+\Delta t}(T) = C\, e^{-a^2(T-t-\Delta t)}. \tag{26}$$

The constant C is determined by the condition that the total WTM area added equals the injected area (25). Integrating (26) over maturity:

$$\int_0^\infty C e^{-a^2\theta}\, d\theta = \frac{C}{a^2} = \Delta A_{\mathrm{WTM}}(t+\Delta t),$$

so that

$$C = a^2 \Delta A_{\mathrm{WTM}}(t+\Delta t) = a^2 \Delta x_{t+\Delta t}\,\Delta t.$$

Therefore the propagated distortion is

$$\boxed{\Delta x_{t+\Delta t}(T) = a^2 \Delta x_{t+\Delta t}\,\Delta t\; e^{-a^2(T-t-\Delta t)}.} \tag{27}$$

The updated observed curve. By adding the policy-induced deformation to the natural expectation (23), we obtain

$$\boxed{x_{\text{obs}}(t+\Delta t, T) = \mathbb{E}[x_{\text{IRFR}}(t+\Delta t, T)] + a^2 \Delta x_{t+\Delta t}\, \Delta t\, e^{-a^2(T-t-\Delta t)}.} \tag{28}$$

Interpretation. The short-rate peg injects a WTM area $\Delta A_{\text{WTM}}(t+\Delta t) = \Delta x_{t+\Delta t}\Delta t$. The IRFR thermodynamic structure forces this area to propagate across maturities through the exponential kernel $e^{-a^2(T-t-\Delta t)}$. The result is a global, smooth deformation of the forward curve—lifting the short end and reducing the amplitude of the expected IRFR deviation. This constitutes the *policy footprint* on the term structure.

1.7 Apparent Drift Suppression and the Onset of Metastability

The short-rate intervention injects a WTM increment $\Delta A_{\text{WTM}}(t) = \Delta x_t\, dt$ at each instant. The observed WTM then evolves as

$$A^{\text{obs}}_{\text{WTM}}(t+dt) = A^{\text{obs}}_{\text{WTM}}(t) e^{-a^2 dt} + \Delta A_{\text{WTM}}(t). \tag{29}$$

Taking the continuous-time limit yields

$$\frac{d}{dt} A^{\text{obs}}_{\text{WTM}}(t) = -a^2 A^{\text{obs}}_{\text{WTM}}(t) + \Delta_{\text{CB}}(t), \qquad \Delta_{\text{CB}}(t) = \Delta x_t. \tag{30}$$

It is convenient to rewrite (30) in relaxation form:

$$\frac{d}{dt} A^{\text{obs}}_{\text{WTM}}(t) = -a^2_{\text{app}}(t)\, A^{\text{obs}}_{\text{WTM}}(t), \qquad a^2_{\text{app}}(t) = a^2 - \frac{\Delta_{\text{CB}}(t)}{A^{\text{obs}}_{\text{WTM}}(t)}. \tag{31}$$

Because $\Delta_{\text{CB}}(t) > 0$, it follows that

$$\boxed{a^2_{\text{app}}(t) < a^2.} \tag{32}$$

Thus persistent intervention suppresses the effective relaxation rate of the system. Drift dissipation becomes too weak to offset the injected WTM area, and the system enters a *metastable regime*. Macroeconomic energy begins to accumulate exponentially. Catastrophic reversion becomes inevitable once $A^{\text{obs}}_{\text{WTM}}(t)$ exceeds the critical capacity A_{crit}.

1.8 The Metastability Theorem and the Time-to-Failure Formula

We now formalise the metastability mechanism induced by persistent short-rate intervention. As shown in the previous subsection, the observed WTM functional evolves under intervention according to

$$\frac{d}{dt} A^{\text{obs}}_{\text{WTM}}(t) = -\, a^2 A^{\text{obs}}_{\text{WTM}}(t) + \Delta_{\text{CB}}(t), \tag{33}$$

where $\Delta_{\text{CB}}(t) > 0$ represents the continuous injection of WTM area (macroeconomic "energy") due to the enforced short-rate peg.

Apparent drift suppression. Equation (33) may be rewritten as

$$\frac{d}{dt}A^{\text{obs}}_{\text{WTM}}(t) = -\,a^2_{\text{app}}(t)\,A^{\text{obs}}_{\text{WTM}}(t), \qquad a^2_{\text{app}}(t) = a^2 - \frac{\Delta_{\text{CB}}(t)}{A^{\text{obs}}_{\text{WTM}}(t)}. \tag{34}$$

Since $\Delta_{\text{CB}}(t) > 0$, it follows that

$$a^2_{\text{app}}(t) < a^2.$$

Define the *drift-suppression gap*

$$\Delta a^2(t) = a^2 - a^2_{\text{app}}(t) = \frac{\Delta_{\text{CB}}(t)}{A^{\text{obs}}_{\text{WTM}}(t)} \;>\; 0. \tag{35}$$

Metastability Theorem. [IRFR Metastability Theorem] Suppose that the central bank injects a continuous stream $\Delta_{\text{CB}}(t) > 0$ of WTM area. If the apparent drift satisfies $a^2_{\text{app}}(t) < a^2$, then the IRFR system enters a metastable regime in which

$$\frac{d}{dt}A^{\text{obs}}_{\text{WTM}}(t) = \Delta a^2(t)\,A^{\text{obs}}_{\text{WTM}}(t), \qquad \Delta a^2(t) > 0.$$

Energy accumulates exponentially, and catastrophic reversion becomes inevitable once $A^{\text{obs}}_{\text{WTM}}(t)$ exceeds A_{crit}.

Time-to-failure. Assume for simplicity that $\Delta a^2(t)$ is constant (the argument extends to bounded time dependence). Then

$$A^{\text{obs}}_{\text{WTM}}(t) = A^{\text{obs}}_{\text{WTM}}(0)\,e^{\Delta a^2 t}. \tag{36}$$

Catastrophic failure occurs when

$$A^{\text{obs}}_{\text{WTM}}(t_{\text{fail}}) = A_{\text{crit}}.$$

Solving gives the time-to-failure:

$$\boxed{t_{\text{fail}} = \frac{1}{\Delta a^2}\ln\left(\frac{A_{\text{crit}}}{A^{\text{obs}}_{\text{WTM}}(0)}\right).} \tag{37}$$

Economic interpretation. The parameter Δa^2 measures the degree of drift suppression induced by intervention. A smaller Δa^2 corresponds to a long metastable period; larger suppression (larger Δa^2) accelerates energy accumulation and reduces t_{fail}. When failure occurs, the system reverts violently on the *natural* timescale a^2, not the suppressed rate a^2_{app}.

1.9 Volatility Humps as Signals of Drift Suppression and Metastability

In the natural (unforced) IRFR system, both the forward curve and its volatility inherit the same one-dimensional exponential kernel. Under identified equilibrium, the diffusion coefficient takes the form

$$g_{\mathrm{IRFR}}(t,T) = a\big(x(t,T) - x_\infty\big) - a x_\infty e^{-a^2(T-t)},$$

so the instantaneous volatility is

$$\sigma_{\mathrm{IRFR}}(t,T) = \sqrt{2}\, g_{\mathrm{IRFR}}(t,T) = C_1(t)\, e^{-a^2(T-t)}.$$

Thus, in the absence of intervention, the volatility term structure is strictly monotone in maturity and cannot exhibit an interior maximum.

Effect of drift suppression. Under short-rate intervention, the short-rate drift is suppressed from its natural thermodynamic value a^2 to a lower apparent value $a^2_{\mathrm{app}} < a^2$. This modifies the evolution of the short end of the yield curve, and therefore the amplitude of the IRFR deviation. In particular, the deviation amplitude evolves on the slower timescale a^2_{app} instead of a^2.

Because the diffusion loading $g(t,T)$ depends both on the instantaneous deviation $x(t,t) - \frac{1}{2}x_\infty$ and on the exponential kernel $e^{-a^2(T-t)}$, drift suppression introduces a second exponential decay rate into the volatility structure. As a result, the observed volatility is no longer governed by a single exponential, but becomes *bi-exponential*:

$$\sigma_{\mathrm{obs}}(t,T) = C_1(t)\, e^{-a^2(T-t)} + C_2(t)\, e^{-a^2_{\mathrm{app}}(T-t)}, \qquad a^2_{\mathrm{app}} < a^2. \tag{38}$$

The origin of the volatility hump. A single exponential function cannot produce an interior maximum. In contrast, a bi-exponential mixture such as (38) exhibits curvature and can generate a hump. Taking the maturity derivative,

$$\frac{d}{d\tau}\sigma_{\mathrm{obs}}(t,\tau) = -\,a^2 C_1(t)\, e^{-a^2\tau} - a^2_{\mathrm{app}} C_2(t)\, e^{-a^2_{\mathrm{app}}\tau}, \qquad \tau = T - t,$$

the hump occurs where the derivative vanishes:

$$a^2 C_1(t)\, e^{-a^2\tau_*} = a^2_{\mathrm{app}} C_2(t)\, e^{-a^2_{\mathrm{app}}\tau_*}. \tag{39}$$

Solving for the hump location yields

$$\tau_* = \frac{1}{a^2_{\mathrm{app}} - a^2} \ln\left(\frac{a^2_{\mathrm{app}} C_2(t)}{a^2 C_1(t)}\right), \tag{40}$$

which is well-defined when $C_1(t)$ and $C_2(t)$ have opposite signs. This is precisely the regime in which drift suppression is active and the energy imbalance forces the volatility curve to bend upward before decaying. Although $C_1 > 0$ in typical non-intervened regimes, rate suppression forces $\Pi(t,T) < 0$, making the policy-induced coefficient C_2 negative; this is exactly the economically relevant regime in which the bi-exponential mixture satisfies $C_1 C_2 < 0$ and produces an interior volatility hump.

Interpretation. The bi-exponential form (38) provides a thermodynamic explanation for volatility humps:

- The $e^{-a^2\tau}$ term represents the natural IRFR thermodynamic decay rate.
- The $e^{-a_{\text{app}}^2\tau}$ term arises from intervention-induced drift suppression.
- The interaction of these two decay rates produces an interior maximum—a volatility hump.

The height of the hump increases with the drift-suppression gap $\Delta a^2 = a^2 - a_{\text{app}}^2$ and with the accumulated WTM energy $A_{\text{WTM}}(t)$. As the system enters a metastable regime, these quantities rise together, causing the hump to steepen sharply. Thus a pronounced hump during periods of apparent yield-curve stability is a natural thermodynamic signal of suppressed drift, latent energy accumulation, and approaching catastrophic reversion.

1.10 Closed–Form Expression for the Volatility Hump Height

Under natural IRFR dynamics, the volatility inherits a single exponential kernel:

$$\sigma_{\text{IRFR}}(t,T) = \sqrt{2}\, C_1(t)\, e^{-a^2(T-t)}.$$

This structure is strictly monotone in maturity and therefore cannot produce an interior maximum.

Under intervention, however, drift suppression modifies the evolution of the short-end state variable $x(t,t)$, reducing its relaxation rate from a^2 to the lower apparent rate $a_{\text{app}}^2 < a^2$. As a result, the volatility inherits *two distinct exponential decay rates.* The observed volatility thus acquires the bi-exponential form

$$\sigma_{\text{obs}}(t,T) = C_1(t)\, e^{-a^2(T-t)} + C_2(t)\, e^{-a_{\text{app}}^2(T-t)}, \qquad a_{\text{app}}^2 < a^2, \tag{41}$$

where

$$C_1(t) \propto x(t,t) - \tfrac{1}{2}x_\infty, \qquad C_2(t) \propto \Delta a^2 A_{\text{WTM}}(t),$$

and $\Delta a^2 = a^2 - a_{\text{app}}^2 > 0$ is the drift-suppression gap.

Hump condition. Let $\tau := T - t$. The volatility curve is

$$\sigma_{\text{obs}}(t,\tau) = C_1(t)\, e^{-a^2\tau} + C_2(t)\, e^{-a_{\text{app}}^2\tau}.$$

The hump occurs when the maturity derivative vanishes:

$$\frac{d}{d\tau}\sigma_{\text{obs}}(t,\tau) = -\,a^2 C_1(t) e^{-a^2\tau} - a_{\text{app}}^2 C_2(t) e^{-a_{\text{app}}^2\tau} = 0.$$

Solving for $\tau = \tau_*$ yields the hump location:

$$\tau_* = \frac{1}{a_{\text{app}}^2 - a^2} \ln\left(\frac{a_{\text{app}}^2 C_2(t)}{a^2 C_1(t)}\right). \tag{42}$$

This interior maximum is possible only because $a_{\text{app}}^2 < a^2$: the bi-exponential structure is essential for hump formation.

Closed–form hump height. Evaluating (41) at $\tau = \tau_*$ gives

$$\sigma_{\text{hump}}(t) = C_1(t)e^{-a^2\tau_*} + C_2(t)e^{-a_{\text{app}}^2\tau_*}.$$

Substituting (42) and simplifying yields the closed-form hump height:

$$\boxed{\sigma_{\text{hump}}(t) = C_1(t)\left(\frac{a^2C_1(t)}{a_{\text{app}}^2C_2(t)}\right)^{\frac{a^2}{a^2-a_{\text{app}}^2}} + C_2(t)\left(\frac{a^2C_1(t)}{a_{\text{app}}^2C_2(t)}\right)^{\frac{a_{\text{app}}^2}{a^2-a_{\text{app}}^2}}.} \tag{43}$$

Using $C_2(t) \propto \Delta a^2 A_{\text{WTM}}(t)$ and $C_1(t) \propto x(t,t) - \frac{1}{2}x_\infty$, this becomes

$$\boxed{\sigma_{\text{hump}}(t) \propto \left(x(t,t) - \tfrac{1}{2}x_\infty\right)^{\frac{a^2}{a^2-a_{\text{app}}^2}} \left(\Delta a^2 A_{\text{WTM}}(t)\right)^{\frac{a_{\text{app}}^2}{a^2-a_{\text{app}}^2}}.} \tag{44}$$

Interpretation. The volatility hump arises from the competition between the natural thermodynamic decay rate a^2 and the suppressed apparent decay rate a_{app}^2. Its height increases with

$$\Delta a^2 = a^2 - a_{\text{app}}^2, \qquad A_{\text{WTM}}(t),$$

so a large hump is direct evidence of drift suppression and accumulated macroeconomic energy. As the system approaches metastability, the bi-exponential mixture steepens, providing a clear early-warning signal of impending catastrophic reversion.

2 Conclusion

This monograph has unified the IRFR representation of the yield curve with a complete thermodynamic framework. WTM and ETM serve as the fundamental macroeconomic invariants governing system dynamics. The IRFR curve emerges as the unique energy–minimizing attractor, while the MSNR law couples drift and volatility into a single structural relationship.

Persistent policy intervention reduces the effective drift from a^2 to a_1, creating a metastable system in which macroeconomic energy accumulates. The Metastability Theorem shows that the time to catastrophic correction is inversely proportional to the drift suppression gap $a^2 - a_1$. This provides a structural explanation for delayed crises and sudden yield–curve blowouts.

The empirical framework developed here enables the identification of intervention, drift suppression, and metastability in real data. The theory offers a powerful diagnostic and predictive toolkit grounded in thermodynamic principles.